# Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

sagargupta.online | Sagargupta16 | sagar-gupta-16-10 | sg85207@gmail.com | +91-8770532413

## Summary

**Cloud Engineer** with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, building multi-account AWS platforms for financial-services and other enterprise customers. Build Terraform landing zones: Organizations OUs with Control Tower, SCPs, tag policies, account baselines, preventative security controls, and network onboarding with Transit Gateway. Terraform module and Landing Zone specialist; 11 upstream PRs merged (axios, Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others). 5x AWS certified including Solutions Architect – Associate.

## Experience

**Amazon Web Services (AWS)** Aug 2024 – Present
*ProServe (Cloud Consultant) - DevOps/MLOps* Hyderabad, India

- **Lead DevOps Consultant, Enterprise customer (Feb 2026 – Present):** Built a Terraform landing zone: Organizations OUs, Control Tower, SCPs, tag policies, account security baselines and least-privilege deploy roles.
  - Automated network onboarding and tag-based firewall access with Transit Gateway.
  - Automated account vending with Terraform Cloud and OIDC federation.
- **DevOps Consultant, Global financial services customer (Oct 2024 – Aug 2025):** Wrote tested Terraform modules (terraform test, PyTest) for a multi-account landing zone. Worked on preventative security controls and EventBridge-based cross-account automation.
- **DevOps Consultant, Financial market infrastructure customer (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns.
- **MLOps Pipeline (SME program, Jun – Aug 2025):** Provisioned end-to-end SageMaker MLOps infrastructure via Terraform + CodePipeline – blue-green endpoints with auto-scaling, CloudFront frontend, API Gateway + Lambda inference, drift-triggered retraining, and alarm-driven automatic rollback. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Internal Impact:** Published two more AWS samples in Terraform, on Organizations governance and a Bedrock AgentCore agent (see Publications). AWS Professional Services MLOps SME program graduate. AWS technical field community ambassador (AI/ML, DevOps, CloudOps, Agentic AI, Serverless); Anthropic Claude Ambassador (2026). Internal tech talks on Terraform testing and GitOps. Associate Speaker Certification (Renewed 2026); Senior Speaker Certification (2026). Security content review qualification (2026). AWS Jam content tester, DevOps and GenAI (2026).

**Amazon Web Services (AWS)** Mar 2024 – Jul 2024
*Associate Professional Services – DevOps (Internship)* Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases with X-Ray and CloudWatch distributed tracing.

**Ikarus-3D** May 2023 – Jul 2023
*Software Developer Associate (Internship)* Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS for continuous code quality analysis.

## Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Published by AWS:** aws-samples/sample-agentcore-harness-agent-terraform (author, MIT-0) – Terraform for a Bedrock AgentCore agent (71 resources in 12 files): MCP tools behind AgentCore Gateway with a default-deny Cedar policy, PROD promotion only after an evaluation passes, CloudWatch alarms, a canary and a budget. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

## Industry Certifications (5)

**AWS**: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
**AWS**: AI Practitioner | Cloud Practitioner

Personal projects and opinions are my own and do not represent Amazon or AWS.

## Projects

**Ledger Sync** Jan 2026
*React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel* GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

**orrery** Oct 2026
*Agent Skill, Claude Code Plugin, Node.js, ELK, SVG* GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

**Blue-Green AWS Terraform** Jul 2025
*Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy* GitHub
◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

**Bedrock Multi-Model MCP** Apr 2026
*TypeScript, MCP Protocol, AWS Bedrock Converse API* GitHub
◇ MCP server putting 26 text models from 7 providers, plus image, video and embedding models, behind 8 tools and one set of AWS credentials. Benchmarked in us-east-1: 26 of 26 models answered and 156 of 156 calls succeeded.

**Claude Cost Optimizer** Mar 2026
*Claude Code Plugin, MCP, TypeScript, React, Python* GitHub | Live
◇ Claude Code plugin, MCP cost server, CLI grader and web calculator for Claude spend. Rebuilt every estimator on cache-aware billing, correcting a modelled 30-turn Opus session from $3.44 to $0.74, and replaced every vague savings range with a sourced or computed number. Golden-value tests pin the cost math in all three implementations.

**Other Notable Projects:** LeetCode Rating Predictor (train/serve skew fix, NumPy serving), itr-agent (deterministic tax engine, 151 tests), farm-stack-aws-ecs (Docker + GitHub Actions OIDC deploy to ECR/ECS), GitScope (Chrome Web Store extension), Instagram Likes Leaderboard (browser-console tool).

## Open Source Contributions (11 merged upstream)

- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

## Education

**National Institute of Technology Warangal, Telangana** Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

**Sanskar College of Professional Studies (DAVV University), Indore** May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

## Technical Skills

**AWS Multi-Account & Governance**: Organizations, OU Design, SCPs, Tag Policies, Landing Zone, Control Tower, AWS Config, Security Hub, IAM Identity Center, Service Catalog, CloudTrail, GuardDuty
**Networking & Hybrid Connectivity**: Transit Gateway (hub-and-spoke), VPC, VPN, Direct Connect, PrivateLink, Route 53 Resolver, Centralized Egress Inspection, Network Firewall
**Compute & Application Services**: ECS Fargate, Lambda, API Gateway, CloudFront, ALB/NLB, EventBridge, Step Functions, SageMaker, Amazon Bedrock AgentCore
**Infrastructure as Code**: Terraform (modules, tftest, PyTest), CloudFormation, HCL, Multi-Account Strategy, Docker
**CI/CD & Automation**: GitHub Actions, CodePipeline, CodeBuild, CodeDeploy, Blue-Green Deployments, GitOps, Policy-as-Code
**Security & Compliance**: Preventative SCPs, KMS, Least-Privilege IAM, Config Rules, Security Hub Standards, AWS Backup, Systems Manager Patch Manager, Amazon Q
**Languages**: HCL, Python, TypeScript, JavaScript, Bash, SQL

Personal projects and opinions are my own and do not represent Amazon or AWS.